Problems 14 - 16
Elastic Net Regression

Uses both L_1 and L_2 penalties together.

L_1 feature selection + L_2 coefficient stability

Objective Function:

$$\hat{\beta}_{EN} = \arg\min_{\beta} \left[\sum_{i=1}^{n} (y_i - x_i'\beta)^2 + \lambda_1 \underbrace{\sum_{j=1}^{p} |\beta_j|}_{\text{LASSO part creates Sparsity}} + \lambda_2 \sum_{j=1}^{p} \beta_j^2 \right]$$

In ML libraries, Elastic Net is often written as:

$$\hat{\beta}_{EN} = \arg\min_{\beta} \left[\underbrace{\frac{1}{2n} \|y - X\beta\|_2^2}_{\text{Avg. Squared Predict. Error}} + \underbrace{\alpha \left(\rho \|\beta\|_1 + \frac{1-\rho}{2} \|\beta\|_2^2 \right)}_{\text{Elastic Net Penalty}} \right]$$

$$\|\beta\|_1 = \sum_{j=1}^{p} |\beta_j| \qquad \|\beta\|_2^2 = \sum_{j=1}^{p} \beta_j^2$$

The parameter ρ controls the mixture b/w LASSO and Ridge.

In sklearn, ρ is called **l1_ratio**

We can see in the above formula,

If $\rho = 1$, Elastic Net becomes pure LASSO
$\rho = 0$, Elastic Net becomes Ridge like

So,
$\alpha \rightarrow$ How hard should coff. be punished?
$\rho \rightarrow$ Should punishment be more LASSO or Ridge-like?

Elastic Net Geometry :

A simplified constraint form is :

$$\rho \sum_{j=1}^{p} |\beta_j| + (1-\rho) \sum_{j=1}^{p} \beta_j^2 \leq c$$

This shape is b/w a diamond and a circle.

l1_ratio $> 0 \Rightarrow$ ElasticNet can do feature selection.

But if l1_ratio is very small, feature elimination becomes much weaker.

On Problem 16, ElasticNet retained 15 of 16 predictors when l1_ratio = 0.05.

Grouping Effect → Suppose two predictors are highly correlated.

ElasticNet can keep both because the Ridge part encourages grouped behaviour.

This is specially useful when variables are economically related.

Let two predictors be almost identical. If they are exactly identical then :

$$x_1 = x_2$$

Their contribution to the prediction is :

$$x_1\beta_1 + x_2\beta_2 = x_1(\beta_1 + \beta_2)$$

Elastic Net KKT Condition

Objective is :

$$\frac{1}{2n} \|y - X\beta\|_2^2 + \alpha\rho \|\beta\|_1 + \frac{\alpha(1-\rho)}{2} \|\beta\|_2^2$$

Let $r = y - X\hat{\beta}$ be the residual

KKT condition says:

First decide whether the variable survives. Then shrink it.

x_j → full column of values for predictor j.

r is also a column

$x_j' r$ → dot product of the two matrices.

If the residual ($r = y - \hat{y}$) is positive, the model predicted too low.
If x_j is high when residuals are positive, then increasing the coefficient on x_j can raise the predictions in exactly those places.

So, the score $\frac{1}{n} x_j' r$ is useful.

Similarly, the negative score is also useful.

If the score is near zero, predictor is not useful.
In this case, x_j is not systematically related to the residuals.
Elastic Net then says that the predictor is not explaining the leftover error. It is not useful enough. That is when the coeff. may be set to zero.

$z_j = \frac{1}{n} x_j' r$ is Usefulness Score

Dividing by n gives us the average. We measure average usefulness per observation.

Elastic Net then compares the usefulness with the penalty.

$$\left| \frac{1}{n} x_j' r \right| \leq \alpha\rho \quad ?$$

If yes: $\hat{\beta}_j = 0$ and the variable is removed.

If no: $\hat{\beta}_j \neq 0$ and the variable survives.

This is the zero condition.

E.g. Problem 16 :

$$\alpha = 0.7565$$
$$\rho = 0.05$$

Variable killing threshold = $\alpha\rho = 0.0378$ (too small)

Ridge Shrinkage force = $\alpha(1-\rho) = 0.7187$

For each coeff. j, the KKT condition is :

$$-\frac{1}{n} x_j' r + \alpha(1-\rho)\hat{\beta}_j + \alpha\rho s_j = 0$$

Where,

$$s_j \in \partial|\hat{\beta}_j|$$

Amount of variable killing depends on :

$$\alpha \times l1_ratio$$

Elastic Net is usually estimated using coordinate descent :

Coord. descent update:

$$\beta_j \leftarrow \frac{S\left(\frac{1}{n} x_j' r, \alpha\rho\right)}{\frac{1}{n} x_j' x_j + \alpha(1-\rho)}$$

The numerator is Soft Thresholding - it decides whether the coeff. survives. The threshold is $\alpha\rho$.

The denominator contains the Ridge part. If predictors are standardised

$$\frac{1}{n} x_j' x_j = 1$$

so the update becomes approx.

$$\beta_j \leftarrow \frac{S\left(\frac{1}{n} x_j' r, \alpha\rho\right)}{1 + \alpha(1-\rho)}$$

Elastic Net = Soft Thresholding + Extra Ridge Shrinkage

If $\hat{\beta}_j = 0$,

then Ridge part disappears.

So, $$-\frac{1}{n} x_j' r + \alpha\rho s_j = 0$$

$$\frac{1}{n} x_j' r = \alpha\rho s_j$$

Since, $s_j \in [-1, 1]$

we get $$\frac{1}{n} x_j' r \in [-\alpha\rho, \alpha\rho]$$

which is the same as :

$$\left| \frac{1}{n} x_j' r \right| \leq \alpha\rho$$

If we recall, Ridge has

$$\hat{\beta}_{Ridge} = (X'X + \lambda I)^{-1} X'y$$

ElasticNet cannot be solved like this because it contains $|\beta_j|$ from the LASSO part.

When updating coeff. j using coordinate descent:

We define $$r_j = y - \sum_{k \neq j} x_k \beta_k$$

This means we remove the contribution of all variables except x_j

The all-zero condition is :

$$\left| \frac{1}{n} x_j' y \right| \leq \alpha\rho \quad \text{for every predictor } j$$

The strongest predictor has :

$$\max_j \left| \frac{1}{n} x_j' y \right|$$

To kill even the strongest predictor, we need:

$$\alpha\rho \geq \max_j \left| \frac{1}{n} x_j' y \right|$$

$$\alpha_{max} = \frac{1}{\rho} \max_j \left| \frac{1}{n} x_j' y \right|$$

This means: The smaller l1_ratio is, the larger α must be to kill all variables.

Elastic Net

- $\alpha\rho$ = selection force
- $\alpha(1-\rho)$ = stabilisation force

KKT Intuition: Elastic Net keeps a predictor only if its correlation with the residual is large enough to overcome the L_1 part of the penalty.

Elastic Net is biased, just like Ridge and LASSO.

Problem 14: $\alpha\rho = 1.3535$

$\alpha(1-\rho) = 0.1504$

Problem 15: $\alpha\rho = 0.2229$

$\alpha(1-\rho) = 0.0248$

Problem 16: $\alpha\rho = 0.037825$

$\alpha(1-\rho) = 0.718675$

The name is Elastic Net because it can stretch b/w LASSO and Ridge-like behaviours.

Elastic Net has 2 main hyperparameters: α, l1_ratio

So, we choose

$$(\alpha^*, \rho^*) = \arg\min_{\alpha,\rho} CV(\alpha, \rho)$$

Choose the combin. of α and l1_ratio that gives the lowest Cross Validn. Error.

The validation design should mimic the actual forecasting problem.
This matters because regularisn. parameters are chosen using validation error.
If validation is unrealistic, alpha and l1_ratio may be misleading.

Elastic Net often produces predictions that are less extreme than OLS. Because coeff. are shrunk.
If coeff. are smaller, predictions tend to move less aggressively. This creates regression towards the mean.

This is good when extreme predictions are noisy. But it is bad if the true target has extreme movements that matter. Problem 16 shows this. This is a typical Regularisation Effect.

Regularisation improves stability, but it may underpredict extremes.